Gmail

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New Entry: Apply to Become a Science Corps Fellow

Science Corps <eduversumdev@gmail.com> Tue, Jun 29, 2021 at 12:51 PM
To: ccho@science-corps.org

Name

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Email

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Gender

Male

City, state, and country of residence

Philadelphia, Pennsylvania, USA

Nationality

USA

When did or will you finish your PhD?

December 2021

University Education

PhD | Bioengineering | University of Pennsylvania, School of Engineering and Applied Science | December 2021

BSE | Bioengineering | University of Pennsylvania, School of Engineering and Applied Science | May 2013
BS | Economics | University of Pennsylvania, The Wharton School | May 2013

What are your reasons for applying to be a Science Corps fellow?

During my graduate studies, I have been fortunate enough to have several opportunities to teach, mentor, and even develop science curriculum for undergraduates at Penn. I found these experiences incredibly rewarding and invigorating; when I was burned out from the grind of academic research, mentoring and teaching others refreshed my love of science. As a mentor, I focused on instilling my mentees with the same child-like curiosity and excitement that first attracted me to science by emphasizing the scientific method as a process by which to understand the world around us, rather than focus solely on the results.

As a Science Corps fellow, I hope to adapt my mentoring experiences to inspire and empower students abroad. I believe that my past experiences will enable me to make genuine connections with the students, and I hope to instill them with the same sense of child-like wonder that I still feel when learning about new science. In addition, I know that working abroad will challenge me as both an educator and scientist, and I want to develop this global perspective as I move into the next phase of my career.

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship.

As a graduate student, I have mentored 20 students, mostly as an advisor to the Penn iGEM team (an undergraduate research team that competes in an international science competition). Through this experience, I have gained skills in project management, leadership, and mentoring while being able to share my passion for science with excited undergraduate students.

As a teaching assistant during my PhD, I co-designed a new lab module teaching synthetic biology for the undergraduate bioengineering curriculum. This involved engineering new biological strains, developing a mathematical model to describe them, and building custom automated imaging hardware. This module has been taught for the last 6 years to over 500 students. From this experience, I gained an appreciation for the difficulty in developing science lab curriculum for large classes. I also practiced teaching in small-group settings which substantially improved my scientific communication skills.

I have also held several leadership positions in the Penn chapter of Engineers without Borders, and I traveled to Cameroon multiples times for implementation projects where we worked closely with a local community NGO. This experience ignited a passion for global service and exposed me to the difficulties, challenges, and rewarding nature of sustainable development.

Where do you imagine yourself professionally in the future? Check all that apply.

Academic scientist in academia
Doing science in industry
In science education
In science outreach

Expand on your selection above and describe where you imagine yourself professionally in the near (two year) and long-term (ten year) future.

In the near term, I would like to continue my academic scientific training through pursuing a post doc. I am primarily looking into non-US research labs as I would like to gain diverse and global training while fulfilling a personal goal of living abroad alone. I have enjoyed the flexibility and freedom associated with academic science, and I hope to continue my training in the short term while develop new cutting-edge research tools for understanding biological systems.

In the long term, I will probably make a shift to industry. Ultimately, I want to work on developing biological technologies that can provide tangible and direct impact to global problems in health or sustainability. I believe biotechnology is at the cusp of an “industrial revolution,” and I am particularly excited about using protein engineering to develop more sustainable manufacturing or food sources.

Regardless of my future role, I hope to find volunteer outlets that allow me to continue to mentor and educate others, whether in schools or community DIYbio labs. I am excited about the future of biotechnology and want to spread this excitement to the broader community.

CV/Resume

Magaraci_CV.pdf

How did you find out about Science Corps?

University email list

Any other comments/questions?

I am flexible in terms of placement, and should be available starting January 2022.

This is an awesome program and opportunity, and the founding story really resonates with some of my own experiences as I have been looking for ways to continue my career development abroad! Thank you for the consideration and I hope we can connect soon!

Sent from Science Corps